

AI2: Information Theory in ML Algorithms

Decision Tree Learning & Feature Selection

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Outline of Topics

- ▶ Decision Tree Learning uses information theoretic quantities
- ▶ Mutual information based Feature Selection

Why Decision Tree Learning?

Decision trees are one of the most popular machine learning algorithms for several reasons:

- ▶ **Simplicity & Interpretability:** Humans can read and use decision trees. [Read this article.](#)
- ▶ **Extendability:** [Random Forests](#), and [Gradient Boosting \(such as XGBoost\)](#) are ensembles of decision trees.

Application examples:

- ▶ [Decision Trees for Credit Card Fraud Detection](#): “It then creates an if/then decision tree for that account of features that do and don’t point to fraud.”
- ▶ [Google Purchases Customer Service Automation Firm Onward](#): “Onward uses a unique decision tree algorithm that allows its bots to be exceptionally functional in delivering world-class customer care.”
- ▶ [AI Reveals The Perfect Pancake Recipe.](#)

Decision Tree Learning

Predictive Model Structure:

- ▶ **Root or Internal Node:** Represents a feature.
- ▶ **Leaf Node:** Represents the target value .
- ▶ **Branch:** Represents a decision rule.

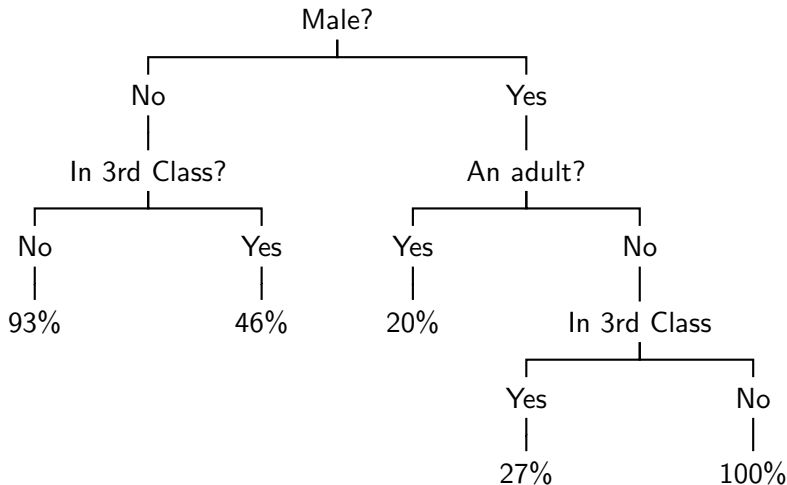
Types of Decision Trees:

- ▶ **Classification Trees:** The target variable takes categorical values (e.g., male/female).
- ▶ **Regression Trees:** The target variable takes continuous values (e.g., temperature).

Popular Algorithms:

- ▶ ID3 (Iterative Dichotomiser 3)
- ▶ C4.5 (successor of ID3)
- ▶ CART (Classification And Regression Tree)

Decision Tree Example: Survival of Titanic Passengers

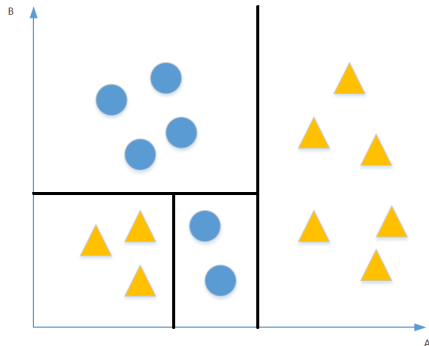


Titanic - Machine Learning from Disaster

How to construct a decision tree?

Given a dataset, the algorithm:

- ▶ Searches for the best rules to split samples that are dissimilar.
- ▶ Repeats the process until the leaves (partitions) reach a certain degree homogeneity.



Gini Index and Information Gain

Question: How do we find the best rules to split the samples?

Answer: Two main methods:

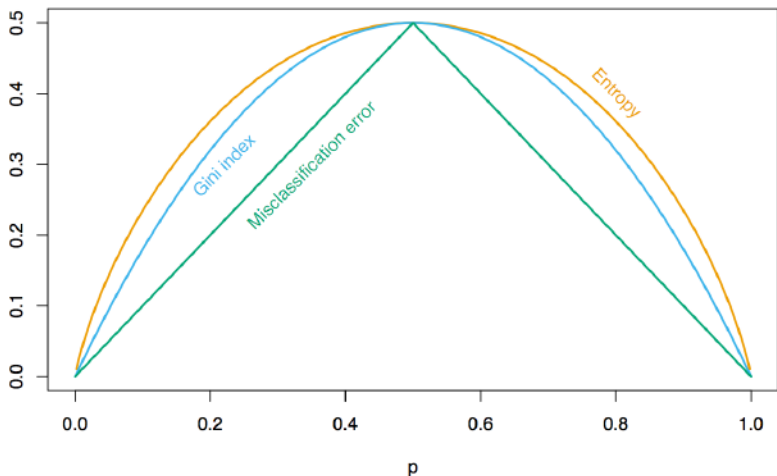
- ▶ **Gini Index (Gini Impurity):** Used in CART (Classification And Regression Trees)

$$I_G(p) = 1 - \sum_{i=1}^J p_i^2$$

where p_i is the fraction of items labeled with class i in the dataset.

- ▶ **Information Gain:** Used in:
 - ▶ ID3 (Iterative Dichotomiser 3)
 - ▶ C4.5 (successor of ID3)

Impurity measures – Illustration



Measures of impurity in 2-class case, as a function of p (the fraction of occurrences of label 1 (or 0)). Here, Entropy is scaled by a factor of 1/2 for the ease of visual comparison.

Information Gain for Decision Trees

Information Gain measures the reduction in entropy after splitting a dataset based on a feature.

$$IG(Y, X) = H(Y) - H(Y|X)$$

where:

- ▶ Y : Random variable representing the target (labels).
- ▶ X : Random variable representing a feature of the input sample.
- ▶ $H(Y)$: Entropy of Y .
- ▶ $H(Y|X)$: Conditional entropy of Y given X .

Interpretation:

- ▶ Information gain quantifies the improvement in classifying labels after using a feature to split the dataset.
- ▶ The feature that maximizes information gain is chosen for the split.

Example: Cancer Diagnosis

Scenario:

- ▶ Biopsies were taken from tissues of 5 patients suspected to have breast cancer.
- ▶ Three biomarkers (A, B, C) were measured as either positive (T) or negative (F).
- ▶ Oncologists confirmed cancer diagnosis for 3 patients, labeled as **T** (True for cancer).

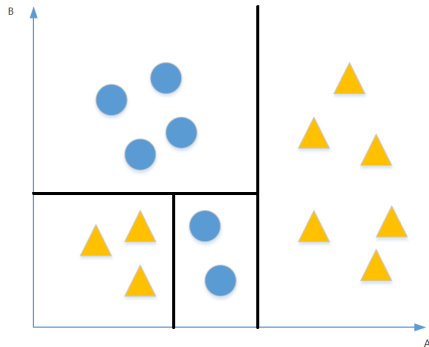
Tissue ID	A	B	C	Label (Cancer?)
1	T	T	T	F
2	T	F	T	T
3	F	F	T	T
4	F	T	T	F
5	F	T	F	T

Question: How do we train a decision tree to split the samples using the features (biomarkers)?

Example: Cancer diagnosis

The target variable Y takes 2 values, we can represent them as 0 (normal) 1 (cancer). Likewise for the input data features ($T=1$, $F=0$). Then we follow the following steps:

Step 1: Calculate the Entropy of the random variable Y before the split.



Example: Cancer Diagnosis - Calculating Entropy

Step 1: Calculate the Entropy of the random variable Y before the split.
Recall its definition:

$$H(Y) = - \sum_{i=1}^n P(Y = y_i) \log P(Y = y_i) = - \sum_{i=1}^n P(y_i) \log P(y_i)$$

Calculating Probabilities:

- ▶ $P(0) = P(\text{Normal}) = \frac{2}{5}$
- ▶ $P(1) = P(\text{Cancer}) = \frac{3}{5}$

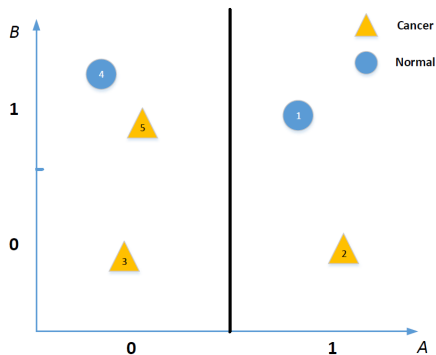
Calculating Entropy:

$$\begin{aligned} H(Y) &= -[P(0) \log P(0) + P(1) \log P(1)] \\ &= -\left(\frac{2}{5} \log \frac{2}{5} + \frac{3}{5} \log \frac{3}{5}\right) \approx 0.971 \end{aligned}$$

Example: Cancer Diagnosis - Conditional Entropy Calculation

Step 2: Calculate the Conditional Entropy after the split, conditioning on each feature in turn. Recall definition of conditional entropy.

Conditioning on feature A:



Example: Cancer Diagnosis - Conditional Entropy Calculation

Step 2: Calculate the Conditional Entropy after the split, conditioning on each feature in turn. Recall definition of conditional entropy.

Conditioning on feature A:

$$\begin{aligned} H(Y|A) &= \sum_i P(A = a_i) H(Y|A = a_i) \\ &= - \sum_i P(A = a_i) \sum_j P(y_j|a_i) \log P(y_j|a_i) \end{aligned}$$

Calculate the probabilities and conditional probabilities

- ▶ $P(A = 0) = \frac{3}{5}$
- ▶ $P(Y = 1|A = 0) = \frac{2}{3}, P(Y = 0|A = 0) = \frac{1}{3}$
- ▶ $P(A = 1) = \frac{2}{5}$
- ▶ $P(Y = 1|A = 1) = \frac{1}{2}, P(Y = 0|A = 1) = \frac{1}{2}$

Calculating the conditional entropy:

$$H(Y|A) = - \left[\frac{3}{5} \left(\frac{2}{3} \log \frac{2}{3} + \frac{1}{3} \log \frac{1}{3} \right) + \frac{2}{5} \left(\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right) \right] \approx 0.951$$

Example: Cancer Diagnosis - Information Gain Calculation

Step 3: Calculate the Information Gain for Split A

Recall the formula definition:

$$IG(Y, A) = H(Y) - H(Y|A)$$

Calculation:

$$\begin{aligned} IG(Y, A) &= 0.971 - 0.951 \\ &= 0.02 \end{aligned}$$

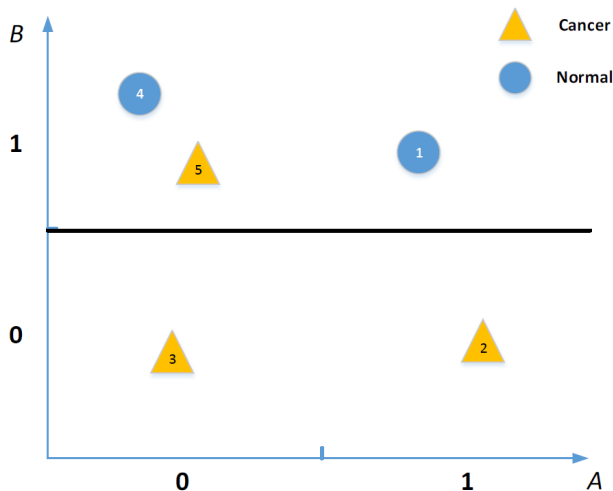
Step 4: Do Steps 1-3 for all the remaining features.

Results of Information Gain obtained:

- ▶ Split on A: $IG(Y, A) = 0.02$
- ▶ Split on B: $IG(Y, B) = 0.419$
- ▶ Split on C: $IG(Y, C) = 0.171$

Conclusion: Split on biomarker B has the highest information gain,

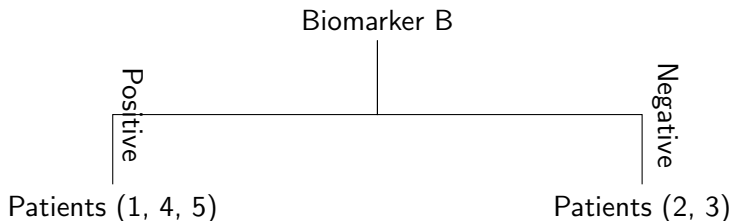
Your first decision tree



Splitting samples based on the values of random variable B generates the largest information gain.

Your First Decision Tree

Splitting Samples Based on Biomarker B



Note:

- ▶ Here, splitting is based directly on the binary values (Positive/Negative) of Biomarker B.
- ▶ For data with real-valued features, thresholds may be used.

Are we done?

Real data sets are typically larger and more complex, so a single split is insufficient to tell apart the classes.

Step 5 Do Steps 1-4 in each branch.

Pause. When to stop growing the tree?

Important: To **prevent overfitting**, we don't want too many leafs, even if the leaf nodes don't achieve 0 entropy. Approaches to guard against overfitting:

- ▶ Early stopping
- ▶ Post-pruning

Limitations of Decision Trees

- ▶ **Unstable:** A small change in the data can lead to a large change in the structure of the obtained decision tree.
- ▶ **Greedy search:** Tree construction has no backtracking, as searching in the space of ALL possible trees is computationally infeasible.
- ▶ **Relatively Inaccurate:** Many other models, such as Support Vector Machines and Neural Networks, often perform better on similar data.

Solutions: Decision Tree Ensembles

- ▶ **Random Forest:** Combines multiple decision trees to reduce overfitting and improve accuracy.
- ▶ **Gradient Boosting (e.g., XGBoost):** Builds an ensemble of trees sequentially, where each tree corrects errors made by previous trees.

Further Reading:

- ▶ [An Introduction to Statistical Learning, Chapters 8.1-8.2](#)

Question: Can we go beyond decision trees?

Information Gain actually is...

Question: What does this equation remind you of?

$$IG(Y, X) = H(Y) - H(Y|X)$$

Information Gain actually is...

Answer: Mutual Information

$$I(X; Y) = H(X) - H(X|Y)$$

Alternative forms include:

$$\begin{aligned} I(X; Y) &= H(Y) - H(Y|X) \\ &= H(X) + H(Y) - H(X, Y) \\ &= H(X, Y) - H(Y|X) - H(X|Y) \\ &= KL(P(X, Y) \| P(X)P(Y)) \end{aligned}$$

Interpretation:

- ▶ Decision tree learning algorithms (e.g., ID3, C4.5) recursively use mutual information to select the feature that shares the most information with the dependent variable.
- ▶ The selected feature is used to split the data.

Mutual Information based Feature Selection

Question: Why not use mutual information to select a set of relevant features before applying a classifier?

Mutual Information Feature Selection:

- ▶ Choose a subset of features S from the initial set F that maximises the mutual information between the target variable Y and the subset of selected features S :

$$\arg \max_{S \subseteq F} I(Y; S)$$

- ▶ But searching through all subsets is computationally too demanding. A greedy approach is feasible instead.

Find out more about feature selection (several other methods exist too):

- ▶ [Wikipedia: Feature Selection](#)
- ▶ [Scikit-learn: Feature Selection Module](#)

Greedy method

Mutual Information Feature Selection Algorithm

1. Initialize: $F \leftarrow$ all features, $S \leftarrow \emptyset$
2. Select the feature $f_{\max}$ that maximizes mutual information:

$$f_{\max} = \arg \max_{X_i \in F} I(Y; X_i)$$

3. Update: $F \leftarrow F \setminus \{f_{\max}\}$, $S \leftarrow S \cup \{f_{\max}\}$
4. **Repeat** until $|S| = K$ (desired number of features):
 - 4.1 Select feature $f_{\max}$ that maximizes:

$$f_{\max} = \arg \max_{X_i \in F} \left(I(Y; X_i) - \beta \sum_{X_s \in S} I(X_s; X_i) \right)$$

- 4.2 Update: $F \leftarrow F \setminus \{f_{\max}\}$, $S \leftarrow S \cup \{f_{\max}\}$

5. End Repeat

Mutual Information based Feature Selection

Further Readings:

- ▶ Hendrik Blockeel, Laurens Devos, Benoit Frénay, Géraldin Nanfack, Siegfried Nijssen: [Decision Trees: From Efficient Prediction to Responsible AI](#), Frontiers in Artificial Intelligence, July 2023.
- ▶ Pablo A. Estevez, Michel Tesmer, Claudio A. Perez, and Jacek M. Zurada: [Normalized Mutual Information Feature Selection](#), IEEE Transactions on Neural Networks, 2009.

The idea of using MI for feature selection has been around for some time:

- ▶ R. Battiti, "Using Mutual Information for Selecting Features," IEEE Transactions on Neural Networks, 1994

Exercise

Task:

- ▶ Implement a decision tree learning algorithm to predict Titanic passenger survival.
- ▶ Apply a mutual information feature selection algorithm and then construct your decision tree.
- ▶ Compare the decision tree learned from data with and without feature selection.

Objective:

- ▶ Evaluate the effectiveness of feature selection on model performance.
- ▶ Understand the role of mutual information in improving decision tree learning.

Additional Resources:

- ▶ [Kaggle: Titanic - Machine Learning from Disaster](#)